

Arjuna JEE 2.0 2027

Maths

Basic Mathematics

DPP: 7

Total Duration - 40 Min

Q1 Find x , if $\log_2 x + \log_4 x + \log_8 x = 11$.

Q2 If $\log_x (\log_{18} (\sqrt{2} + \sqrt{8})) = \frac{1}{3}$, Then the value of $1000x$ is equal to

- (A) 8 (B) $1/8$
(C) $1/125$ (D) 125

Q3 If $\log_4 5 = x$ and $\log_5 6 = y$, then $\log_3 2$ is equal to

- (A) $\frac{1}{2x+1}$
(B) $\frac{1}{2y+1}$
(C) $2xy + 1$
(D) $\frac{1}{2xy-1}$

Q4 If $\log(x - y) - \log 5 - \frac{1}{2}\log x - \frac{1}{2}\log y = 0$ then

$$\frac{x}{y} + \frac{y}{x} =$$

- (A) 25 (B) 26
(C) 27 (D) 28

Q5 The numerator and denominator of a fraction are in ratio 2 : 3. If 6 is subtracted from the numerator, the result will be $\frac{2}{3}$ of the original fraction. What is the value of the numerator?

- (A) 27 (B) 18
(C) 36 (D) 6

Q6 Arrange the fractions in ascending order.

- (A) $\frac{11}{14}, \frac{16}{19}, \frac{19}{21}$
(B) $\frac{16}{19}, \frac{11}{14}, \frac{19}{21}$
(C) $\frac{19}{21}, \frac{11}{14}, \frac{16}{19}$
(D) $\frac{16}{19}, \frac{19}{21}, \frac{11}{14}$

Q7 If x, y, z are all different real numbers, then prove

$$\text{that } \frac{1}{(x-y)^2} + \frac{1}{(y-z)^2} + \frac{1}{(z-x)^2} = \left(\frac{1}{x-y} + \frac{1}{y-z} + \frac{1}{z-x} \right)^2$$

Q8 If $\frac{a}{b} = \frac{2}{3}$ and $\frac{b}{c} = \frac{4}{5}$, then find value of $\frac{a+b}{b+c}$

Q9 The sum of the 1st 50 natural numbers is.

- (A) 1275 (B) 1325
(C) 1450 (D) 1375

Q10 If $x = 3 + \sqrt{8}$ and $y = 3 - \sqrt{8}$ then

$$\frac{1}{x^2} + \frac{1}{y^2} =$$

- (A) -34 (B) 34
(C) 12 (D) -12

Q11 If $\frac{3+\sqrt{7}}{3-\sqrt{7}} = a + b\sqrt{7}$ then $(a, b) =$

- (A) $(8, -3)$ (B) $(-8, -3)$
(C) $(-8, 3)$ (D) $(8, 3)$

Q12 If sum of two numbers is C and their quotient is $\frac{P}{Q}$ find the numbers.

Q13 If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f}$, then find the value of $\frac{2a^4b^2+3a^2c^2-5e^4f}{2b^6+3b^2d^2-5f^5}$ in terms of a and b .

Q14 If $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2)$, Show that $= (ax + by + cz)^2$
 $x : a = y : b = z : c$.

Q15 If $x^2 + 5y^2 + z^2 - 4xy + 2yz = 0$, $x, y, z \neq 0, x, y, z \in R$, then

- (A) $\frac{x}{y} = 2$
(B) $\frac{x}{y} + \frac{y}{z} = 1$
(C) $\frac{x}{y} - \frac{y}{z} = 3$
(D) $\frac{x}{z} = -2$

Q16 If a, b, c are non-zero real numbers, then the minimum value of expression



$$\left(\frac{(a^4 + 3a^2 + 1)(b^4 + 5b^2 + 1)(c^4 + 7c^2 + 1)}{a^2 b^2 c^2} \right)$$

is

Q17 Integral value of x which satisfies the equation
 $\log_6 54 + \log_x 16 = \log_{\sqrt{2}} x - \log_{36} (4/9)$
 is _____

Q18 If
 $x, y, z \in \mathbb{R}, x^2 + 4y^2 + 9z^2 - 2x - 4y - 6z$ then
 $+ 3 = 0$
 value of $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ equals
 (A) 1 (B) 6
 (C) 2 (D) cannot say

Q19 $\left(\frac{x^b}{x^c}\right)^{b+c-a} \left(\frac{x^c}{x^a}\right)^{c+a-b} \left(\frac{x^a}{x^b}\right)^{a+b-c}$ equals
 (A) x^{abc} (B) 1
 (C) $x^{(ab+bc+ca)}$ (D) x^{a+b+c}

Q20 If $\frac{(2^{n+1})^m \cdot 2^{2n} \cdot 2^n}{(2^{m+1})^n \cdot 2^{2m}} = 1$ then m is equal to
 (A) 0 (B) 1
 (C) n (D) $2n$

Q21 a, b, c are real numbers such that $a + b + c = 3$
 and

$$\frac{1}{a+b} + \frac{1}{b+c} + \frac{1}{c+a} = \frac{10}{3}.$$

The value of $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}$ is



Answer Key

Q1 64

Q2 (D)

Q3 (D)

Q4 (C)

Q5 (B)

Q6 (A)

Q7 Refer video sol.

Q8 $\frac{20}{27}$

Q9 (A)

Q10 (B)

Q11 (D)

Q12 $\frac{CP}{P+Q}, \frac{CQ}{P+Q}$ Q13 $\frac{a^4}{b^4}$

Q14 proof

Q15 (A, B, C, D)

Q16 (A)

Q17 4

Q18 (B)

Q19 (B)

Q20 (D)

Q21 7

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